

ECE 330 Exam 1: Spring 2023
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 90 Minutes

NAME _____

NetID (please print) _____

USEFUL INFORMATION

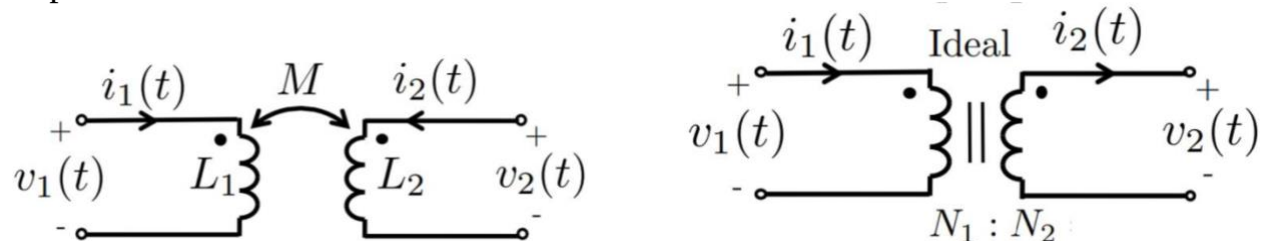
$$\begin{array}{lll} \sin(x) = \cos(x - 90^\circ) & \bar{V} = \bar{Z}\bar{I} & \bar{S} = \bar{V}\bar{I}^* = P + jQ & \bar{S}_{3\phi} = \sqrt{3}V_L I_L \angle \theta \\ 0 < \theta < 180^\circ \text{ (lag)} & I_L = \sqrt{3}I_\phi \text{ (delta)} & \bar{Z}_Y = \bar{Z}_\Delta / 3 & \\ -180^\circ < \theta < 0 \text{ (lead)} & V_L = \sqrt{3}V_\phi \text{ (wye)} & \mu_0 = 4\pi \times 10^{-7} \text{ H/m} & \end{array}$$

ABC phase sequence has A at 0°, B at -120°, and C at +120°

$$\int \underline{H} \cdot \underline{dl} = \int \underline{J_f} \cdot \hat{n} dA \quad \int \underline{E} \cdot \underline{dl} = -\frac{d}{dt} \left(\int \underline{B} \cdot \hat{n} dA \right) \quad \mathcal{R} = \frac{l}{\mu A} \quad Ni = \mathcal{R}\phi$$

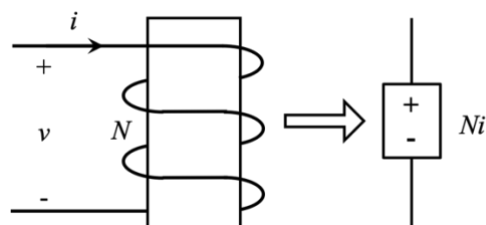
$$\phi = BA \quad \lambda = N\phi = Li \text{ (if linear)} \quad v = \frac{d\lambda}{dt} \quad k = \frac{M}{\sqrt{L_1 L_2}}$$

$$1 \text{ hp} = 746 \text{ W}$$



$$\begin{aligned} v_1 &= L_1 \frac{di_1}{dt} + M \frac{di_2}{dt} \\ v_2 &= M \frac{di_1}{dt} + L_2 \frac{di_2}{dt} \end{aligned}$$

$$\begin{aligned} a &= \frac{N_1}{N_2} & N_1 i_1 &= N_2 i_2 \\ \frac{v_1}{v_2} &= \frac{N_1}{N_2} \end{aligned}$$



Problem 1 (25 Points)

A $2\angle 0^\circ$ kV, 60 Hz supply powers 2 single phase loads connected in parallel. The total current supplied to the 2 loads is 100 A at a lagging power factor. Both loads consume a total power of 150 kW, and load 1 consumes 80 kVA at a PF=0.95 lagging.

- a) What is the angle of the current supplied to the 2 loads?

$\theta_i =$ _____

- b) What is the complex power consumed by load 2?

$\bar{S}_2 =$ _____

- c) What is the impedance of a capacitive load that needs to be added in parallel to the 2 loads to achieve an overall power factor of PF=0.975 lagging? (You may state your answer either in rectangular or polar form)

$$\overline{Z_{cap}} = \underline{\hspace{15cm}}$$

Problem 2 (25 Points)

A balanced 3-phase, 3 wire, 60 Hz, wye-connected source is serving a wye-connected load of unknown total power.

The three-phase line voltage and line current measured at the source are 578 V and 30 A respectively. You may ignore the impedance of the wires connecting the source to the load.

- a) What is the per-phase voltage and per-phase current magnitudes measured at the unknown wye-connected load?

$V_{\phi} =$ _____
 $I_{\phi} =$ _____

In attempt to reduce the line current at the source, 30 kVAR of 3-phase delta-connected capacitance is added in parallel to the loads

- b) What is the per-phase voltage and per-phase current magnitudes measured at the capacitor?

$V_{\phi} =$ _____
 $I_{\phi} =$ _____

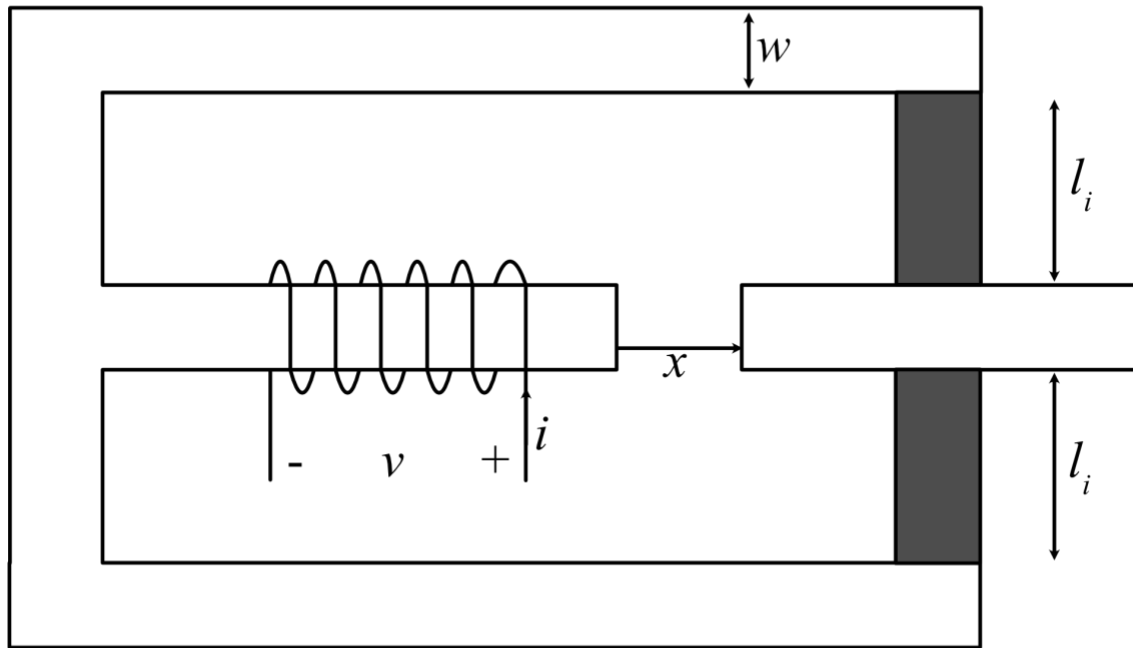
With the capacitance added, the line current at the source *increases* to 45 A, instead of decreasing as we had intended.

- c) What is the total three-phase complex power of the unknown wye-connected load? Please take care to indicate whether the load is capacitive or inductive, and state your units. (You may state your answer either in rectangular or polar form)

CAPACITIVE / INDUCTIVE

$\bar{S}_{Load} =$ _____

Problem 3 (25 Points)



An infinitely permeable iron core given above with width $w=2$ cm and depth into the page $d=3$ cm has a coil with 450 turns wrapped around it as shown. The core contains two inserts with a permeability of $\mu_i=2450\mu_0$ with a length $l_i=10$ cm. The width of the inserts and the depth into the page of the inserts are also $w=2$ cm and $d=3$ cm. The core also contains an air gap with $x=20$ mm. Fringing effects can be neglected.

- a) Draw the magnetic equivalent circuit for this system. Include flux direction, MMF polarity, and numerical values for reluctances.

- b) What is the numerical value for the total flux linkage of the coil written in terms of the unknown current $i(t)$?

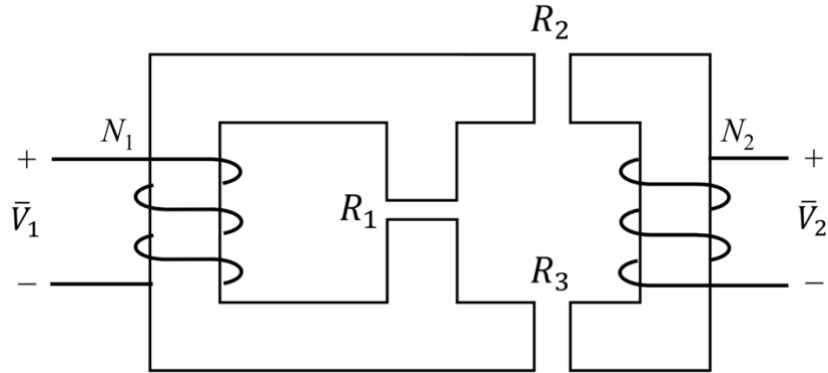
$\lambda =$ _____

- c) If the current into the coil is given as a function of time as $i(t) = 5(1 - e^{-4t})$, what is the numerical value for the voltage across the coil $v(t)$ as a function of time t ?

$v(t) =$ _____

Problem 4 (25 Points)

Your firm is building a transformer, comprising two coils that are wound to have $N_1 = 400$ turns and $N_2 = 10$ turns respectively. The three air gaps are sized to have reluctances of $R_1 = 10 \text{ At/Wb}$, $R_2 = 500 \text{ At/Wb}$, and $R_3 = 500 \text{ At/Wb}$ as labeled. You may treat the permeability of the magnetic core as infinite.



- a) Assuming for now that the transformer is ideal, what would the output voltage $\bar{V}_2$ be at open circuit with an input voltage $\bar{V}_1 = 1\angle 45^\circ \text{ kV (rms)}$ at 60 Hz? (Please specify both the phase and the magnitude of the output voltage $\bar{V}_2$, noting that the magnitude must be a nonnegative number, and briefly justify your answer.)

$$\bar{V}_2 = \underline{\hspace{10cm}}$$

- b) Compute the self inductances L_1 , L_2 , and the mutual inductance M of the two coils.

$L_1 =$ _____
 $L_2 =$ _____
 $M =$ _____

- c) Using the self and mutual inductance values you have just computed, what is the actual output voltage $\bar{V}_2$ at open circuit with an input voltage $\bar{V}_1 = 1\angle 45^\circ$ kV (rms) at 60 Hz? (Please specify both the phase and the magnitude of the output voltage $\bar{V}_2$, noting that the magnitude must be a nonnegative number, and briefly justify your answer.)

$\bar{V}_2 =$ _____